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Inventor(s)	Suzuki; Hiroshi

Sliding component

Abstract

Provided is a sliding component capable of stably reducing the frictional resistance between sliding surfaces entailing eccentric rotation. A sliding component has a sliding surface relatively sliding with eccentric rotation. The sliding surface is provided with a dynamic pressure generation groove defined by side walls extending in a circumferential direction. At least one of the side walls is formed in a waveshape with amplitude in a radial direction.

Inventors:	Suzuki; Hiroshi (Tokyo, JP)
Applicant:	EAGLE INDUSTRY CO., LTD. (Tokyo, JP)
Family ID:	79552531
Assignee:	EAGLE INDUSTRY CO., LTD. (N/A, JP)
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Field of Classification Search

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Primary Examiner: Byrd; Eugene G

Attorney, Agent or Firm: HAYES SOLOWAY P.C.

Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a sliding component used in a rotating machine including an eccentric mechanism.

BACKGROUND ART

(2) Machines entailing rotational driving and used in various industrial fields include not only a rotating machine rotating with its central axis held at a fixed position but also a rotating machine rotating with its central axis entailing eccentricity. The rotating machine rotating with eccentricity is, for example, a scroll compressor. This type of compressor is provided with, for example, a scroll compression mechanism including a fixed scroll having a spiral lap on the surface of an end plate and a movable scroll having a spiral lap on the surface of an end plate and an eccentric mechanism eccentrically rotating a rotary shaft. By the rotary shaft rotating, the movable scroll is slid relative to the fixed scroll with eccentric rotation. As a result, in this mechanism, the fluid supplied from the low-pressure chamber on the outer diameter side of the two scrolls is pressurized and a high-pressure fluid is discharged from the discharge hole formed in the middle of the fixed scroll.

(3) These scroll compressors using the mechanism in which the movable scroll is slid relative to the fixed scroll with eccentric rotation are widely used in, for example, refrigeration cycles because the compressors are highly efficient in terms of compression and cause little noise. However, the compressors are problematic in that a refrigerant leaks from the axial gap between the two scrolls. The scroll compressor described in Patent Citation 1 includes a thrust plate sliding relative to a

movable scroll on the back surface side of the movable scroll. A part of a refrigerant compressed by a scroll compression mechanism is supplied to the back pressure chamber formed on the back surface side of the thrust plate, and the movable scroll is pressed toward a fixed scroll. As a result, it is possible to reduce the leakage of the refrigerant from the axial gap between the two scrolls when the refrigerant is compressed.

CITATION LIST

Patent Literature

(4) Patent Citation 1: JP 2016-61208 A (Pages 5 to 6, FIG. 1)

SUMMARY OF INVENTION

Technical Problem

(5) In the scroll compressor described in Patent Citation 1, a part of the refrigerant compressed by the scroll compression mechanism is used to press the movable scroll from the back surface side toward the fixed scroll via the thrust plate. Accordingly, although the refrigerant leakage from the inter-scroll axial gap can be reduced, a pressing force acts from both axial sides between the two scrolls, especially on the sliding surface that entails the eccentric rotation between the movable scroll and the thrust plate. As a result, there is a problem that an increase in frictional resistance occurs, a smooth operation of the movable scroll is hindered, and the efficiency of compression cannot be enhanced.

(6) The present invention has been made in view of such a problem, and an object of the present invention is to provide a sliding component capable of stably reducing the frictional resistance between sliding surfaces entailing eccentric rotation.

Solution to Problem

(7) In order to solve the above problem, a sliding component according to the present invention is a sliding component having a sliding surface relatively sliding with eccentric rotation, wherein the sliding surface is provided with a dynamic pressure generation groove defined by side walls extending in a circumferential direction, at least one of the side walls being formed in a waveshape with amplitude in a radial direction. According to the aforesaid feature of the present invention, the side wall of the dynamic pressure generation groove extends in the circumferential direction in a waveshape having an amplitude in the radial direction. Accordingly, it is possible to ensure a plurality of intersection surfaces generating dynamic pressure regardless of the eccentric rotation angle. As a result, the sliding surface can be separated from another sliding surface in a state where the inclination thereof is reduced, and the frictional resistance between the sliding surfaces can be stably reduced.

(8) It may be preferable that both the side walls of the dynamic pressure generation groove are formed in the waveshape with the amplitude in the radial direction. According to this preferable configuration, when the sliding surface relatively slides with eccentric rotation, dynamic pressure can be generated at a plurality of points in the circumferential direction on both the side wall on the inner diameter side and the side wall on the outer diameter side. Accordingly, the sliding surface is separated in a state where the inclination thereof is reduced.

(9) It may be preferable that the dynamic pressure generation groove is formed in an annular shape. According to this preferable configuration, dynamic pressure can be generated over the entire circumference of the sliding surface.

(10) It may be preferable that the waveshape of the side wall is periodically formed with a constant pitch. According to this preferable configuration, dynamic pressure can be equally generated over the circumferential direction of the sliding surface.

(11) It may be preferable that the waveshape of the side wall is formed by curved surfaces. According to this preferable configuration, the movement of the fluid that moves in the dynamic pressure generation groove becomes smooth and dynamic pressure can be stably generated.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a schematic configuration diagram illustrating a scroll compressor to which a side seal as a sliding component according to a first embodiment of the present invention is applied.
- (2) FIG. 2 is a diagram illustrating a sliding surface of the side seal in the first embodiment of the present invention.
- (3) FIG. 3 is an A-A cross-sectional view.
- (4) FIG. 4 is a diagram illustrating relative sliding between the sliding surface of the side seal and a sliding surface of a thrust plate in the first embodiment of the present invention. It should be noted that FIG. 4A illustrates the start position of the relative sliding and FIGS. 4B to 4D illustrate the positional relationships between the relatively sliding surfaces of the side seal and the thrust plate at an eccentric rotary shaft rotation of 90, 180, and 270 degrees, respectively.
- (5) FIG. 5 is a diagram illustrating where the dynamic pressure generated in a dynamic pressure generation groove is generated on the sliding surface of the side seal eccentrically rotating from the state of FIG. 4A toward the state of FIG. 4B.
- (6) FIG. 6 is a diagram illustrating a sliding surface of a side seal as a sliding component according to a second embodiment of the present invention.
- (7) FIG. 7 is a diagram illustrating a sliding surface of a side seal as a sliding component according to a third embodiment of the present invention.

DESCRIPTION OF EMBODIMENTS

- (8) Modes for implementing the sliding component according to the present invention will be described below based on embodiments.

First Embodiment

- (9) The sliding component according to a first embodiment of the present invention will be described with reference to FIGS. 1 to 5. It should be noted that in the drawings, the groove formed in the sliding surface of the sliding component or the like is dotted for convenience of description.
- (10) The sliding component according to the first embodiment of the present invention is applied to a rotating machine including an eccentric mechanism such as a scroll compressor C that suctions, compresses, and discharges a refrigerant as a fluid used in the air conditioning system of an automobile or the like. It should be noted that the refrigerant in the present embodiment is a gas mixed with a mist-like lubricating oil.
- (11) First, the scroll compressor C will be described. As illustrated in FIG. 1, the scroll compressor C mainly includes a housing 1, a rotary shaft 2, an inner casing 3, a scroll compression mechanism 4, a side seal 7 as a sliding component, a thrust plate 8, and a drive motor M.
- (12) The housing 1 includes a cylindrical casing 11 and a cover 12 blocking an opening of the casing 11. The drive motor M blocks the opening in the casing 11 that is on the side axially opposite to the opening in the casing 11 that is blocked by the cover 12.
- (13) Formed in the casing 11 are a low-pressure chamber 20, a high-pressure chamber 30, and a back pressure chamber 50. A low-pressure refrigerant is supplied from a refrigerant circuit (not illustrated) to the low-pressure chamber 20 as the external space on the low-pressure side through a suction port 10. A high-pressure refrigerant compressed by the scroll compression mechanism 4 is discharged to the high-pressure chamber 30. A part of the refrigerant compressed by the scroll compression mechanism 4 is supplied, together with lubricating oil, to the back pressure chamber 50 as the external space on the high-pressure side. It should be noted that the back pressure chamber 50 is formed in the cylindrical inner casing 3 accommodated in the casing 11.
- (14) A discharge communication passage 13 is formed in the cover 12. The discharge communication passage 13 allows the refrigerant circuit (not illustrated) and the high-pressure chamber 30 to communicate with each other. In addition, a part of a back pressure communication

passage **14** for communication between the high-pressure chamber **30** and the back pressure chamber **50** is formed in the cover **12** by branching off from the discharge communication passage **13**. It should be noted that the discharge communication passage **13** is provided with an oil separator **6** for lubricating oil separation from a refrigerant.

(15) The inner casing **3** is fixed with an axial end portion of the inner casing **3** abutting against an end plate **41a** of a fixed scroll **41** constituting the scroll compression mechanism **4**. In addition, the inner casing **3** has a side wall where a suction communication passage **15** penetrating the wall in the radial direction is formed. In other words, the low-pressure chamber **20** is formed from the outside of the inner casing **3** to the inside of the inner casing **3** via the suction communication passage **15**. The refrigerant supplied to the inside of the inner casing **3** through the suction communication passage **15** is suctioned into the scroll compression mechanism **4**.

(16) The scroll compression mechanism **4** mainly includes the fixed scroll **41** and a movable scroll **42**. The fixed scroll **41** is fixed to the cover **12** in a sealed shape. The movable scroll **42** is accommodated in the inner casing **3**.

(17) The fixed scroll **41** is made of metal and includes a spiral lap **41b**. The spiral lap **41b** projects toward the movable scroll **42** from the surface of the disk-shaped end plate **41a**, that is, the end plate **41a**. In addition, the fixed scroll **41** has a recessed portion **41c** where the inner diameter side of the back surface of the end plate **41a**, that is, the end surface of the end plate **41a** that abuts against the cover **12** is recessed in the direction opposite to the cover **12**. The high-pressure chamber **30** is defined from the recessed portion **41c** and the cover **12**.

(18) The movable scroll **42** is made of metal and includes a spiral lap **42b**. The spiral lap **42b** projects toward the fixed scroll **41** from the surface of a disk-shaped end plate **42a**, that is, the end plate **42a**. In addition, a boss **42c** protruding from the middle of the back surface of the end plate **42a** is formed on the movable scroll **42**. An eccentric portion **2a** formed on the rotary shaft **2** is fitted into the boss **42c** so as to be relatively rotatable. It should be noted that an eccentric mechanism causing the rotary shaft **2** to perform eccentric rotation is configured by the eccentric portion **2a** of the rotary shaft **2** and a counterweight portion **2b** protruding in the outer diameter direction from the rotary shaft **2** in the present embodiment.

(19) When the rotary shaft **2** is rotationally driven by the drive motor **M**, the eccentric portion **2a** rotates eccentrically and the movable scroll **42** slides, in a posture-maintained state, relative to the fixed scroll **41** with the eccentric rotation. At this time, the movable scroll **42** rotates eccentrically with respect to the fixed scroll **41** and, with this rotation, the contact positions of the laps **41b** and **42b** sequentially move in the rotation direction and a compression chamber **40** formed between the laps **41b** and **42b** gradually shrinks while moving toward the middle. As a result, the refrigerant suctioned into the compression chamber **40** from the low-pressure chamber **20** formed on the outer diameter side of the scroll compression mechanism **4** is compressed and, finally, the high-pressure refrigerant is discharged to the high-pressure chamber **30** through a discharge hole **41d** provided in the middle of the fixed scroll **41**.

(20) Next, the side seal **7** as a sliding component in the present embodiment will be described. As illustrated in FIGS. **2** and **3**, the side seal **7** is made of resin and has a rectangular cross section and an annular shape in an axial view. In addition, the side seal **7** is fixed to the back surface of the end plate **42a** of the movable scroll **42** (see FIG. **1**). It should be noted that a sliding surface **7a** of the side seal **7** is illustrated in FIG. **2**.

(21) The side seal **7** has the sliding surface **7a** abutting against a sliding surface **8a** (see FIG. **1**) of the thrust plate **8**.

(22) As illustrated in FIG. **2**, the sliding surface **7a** of the side seal **7** includes a dynamic pressure generation groove **70** formed in an endless ring shape along the sliding surface **7a**.

(23) As illustrated in FIGS. **2** and **3**, the dynamic pressure generation groove **70** is partitioned by an inner side wall **70a**, an outer side wall **70b**, and a bottom surface **70c**. The inner side wall **70a** is formed as a side wall on the inner diameter side extending in the depth direction orthogonal to a

flat surface **72a** of a land **72** on the inner diameter side. The outer side wall **70b** is formed as a side wall on the outer diameter side extending in the depth direction orthogonal to a flat surface **73a** of a land **73** on the outer diameter side. The bottom surface **70c** extends parallel to the surfaces **72a** and **73a** formed on the same surface and is formed by connecting the end portions of the inner side wall **70a** and the outer side wall **70b**.

(24) The inner side wall **70a** and the outer side wall **70b** have a waveshape extending in the circumferential direction and having an amplitude in the radial direction when viewed from the axial direction. The inner side wall **70a** and the outer side wall **70b** have a waveshape with a constant period and a constant amplitude.

(25) It should be noted that in the present embodiment below, the radial apexes in the amplitude of the inner side wall **70a** are referred to as an inner diameter side apex **74** and an outer diameter side apex **75**, and the radial apexes in the amplitude of the outer side wall **70b** are referred to as an inner diameter side apex **76** and an outer diameter side apex **77**.

(26) Further, it should be noted that the part extending in an S shape from the inner diameter side apex **74** of the inner side wall **70a** toward the outer diameter side apex **75** adjacent to the outer diameter side in the clockwise direction with respect to the middle of the side seal **7** is referred to as an intersection surface **701**. In addition, the part extending in an inverted S shape from the inner diameter side apex **74** toward the outer diameter side apex **75** adjacent to the outer diameter side in the counterclockwise direction with respect to the middle of the side seal **7** is referred to as an intersection surface **702**. In addition, the part extending in an S shape from the inner diameter side apex **76** of the outer side wall **70b** toward the outer diameter side apex **77** adjacent to the outer diameter side in the clockwise direction with respect to the middle of the side seal **7** is referred to as an intersection surface **703**. In addition, the part extending in an inverted S shape from the inner diameter side apex **76** toward the outer diameter side apex **77** adjacent to the outer diameter side in the counterclockwise direction with respect to the middle of the side seal **7** is referred to as an intersection surface **704**.

(27) Specifically, as illustrated in FIG. 2, the inner diameter side apexes **74**, the outer diameter side apexes **75**, the inner diameter side apexes **76**, and the outer diameter side apexes **77** each are equally arranged in the circumferential direction. As for the inner diameter side apex **74** and the outer diameter side apex **75** adjacent to each other on the inner side wall **70a**, a maximum amplitude **L1a** on the inner diameter side and a maximum amplitude **L1b** on the outer diameter side are equal with respect to a virtual reference line α extending on the circumference, and these maximum amplitudes **L1a** and **L1b** are equal over the circumferential direction.

(28) Further, as for the inner diameter side apex **76** and the outer diameter side apex **77** adjacent to each other on the outer side wall **70b**, a maximum amplitude **L2a** on the inner diameter side and a maximum amplitude **L2b** on the outer diameter side are equal with respect to a virtual reference line β extending on the circumference, and these maximum amplitudes **L2a** and **L2b** are equal over the circumferential direction. It should be noted that the maximum amplitudes **L1a** and **L1b** and the maximum amplitudes **L2a** and **L2b** have the same length (i.e., $L1a, L1b=L2a, L2b$).

(29) In addition, a circumferential distance **L3** (that is, wavelength) between the inner diameter side apexes **74** adjacent on the inner side wall **70a** is constant over the circumferential direction, and a circumferential distance **L4** (that is, wavelength) between the inner diameter side apexes **76** adjacent on the outer side wall **70b** is constant over the circumferential direction.

(30) In addition, the distance **L3** is substantially equal to the distance **L4**. Strictly speaking, the distance **L4** is slightly longer than the distance **L3** in terms of length ($L3=L4$). It should be noted that although not illustrated, the circumferential distance between the outer diameter side apexes **75** is the same length as the distance **L3** and the circumferential distance between the outer diameter side apexes **77** is the same length as the distance **L4**.

(31) In addition, the inner side wall **70a** and the outer side wall **70b** have a waveshape having a curved surface shape. Specifically, the part near the inner diameter side apex **74** and the inner

diameter side apex **76** is a curved surface convex to the inner diameter side, and the part near the outer diameter side apex **75** and the outer diameter side apex **77** is a curved surface convex to the outer diameter side.

(32) As illustrated in FIG. 3, a radial width dimension L5 of the dynamic pressure generation groove **70** (that is, the separation width between the inner side wall **70a** and the outer side wall **70b**) is constant over the circumferential direction and formed larger than a depth dimension L6 of the dynamic pressure generation groove **70** (i.e., $L5 > L6$). The radial width dimension L5 is constant over the entire circumference of the dynamic pressure generation groove **70**. It should be noted that the width dimension L5 is preferably 10 times or more the depth dimension L6 although the width dimension and the depth dimension of the dynamic pressure generation groove **70** can be freely changed on condition that the width dimension of the dynamic pressure generation groove **70** is formed larger than the depth dimension of the dynamic pressure generation groove **70**. It should be noted that for convenience of description, the depth dimension L6 of the dynamic pressure generation groove **70** illustrated in FIG. 3 is deeper than the actual depth dimension L6.

(33) Referring to FIG. 1, the thrust plate **8** is made of metal and has an annular shape. In addition, a seal ring **43** is fixed to the thrust plate **8**. The seal ring **43** abuts against the inside surface of the inner casing **3**. As a result, the thrust plate **8** functions as a thrust bearing that receives an axial load of the movable scroll **42** via the side seal **7**.

(34) In addition, the side seal **7** and the seal ring **43** partition the low-pressure chamber **20** formed on the outer diameter side of the movable scroll **42** and the back pressure chamber **50** formed on the back surface side of the movable scroll **42** in the inner casing **3**. The back pressure chamber **50** is a closed space formed between the inner casing **3** and the rotary shaft **2**. A seal ring **44** is fixed to the inner periphery of a through hole **3a** provided in the middle of the other end of the inner casing **3** and is in sliding contact in a sealed shape with the rotary shaft **2** inserted through the through hole **3a**. In addition, the back pressure communication passage **14** allowing the high-pressure chamber **30** and the back pressure chamber **50** to communicate with each other is formed over the cover **12**, the fixed scroll **41**, and the inner casing **3**. In addition, the back pressure communication passage **14** is provided with an orifice (not illustrated) and, after depressurization adjustment by means of the orifice, the refrigerant in the high-pressure chamber **30** is supplied to the back pressure chamber **50** together with the lubricating oil separated by the oil separator **6**. At this time, the pressure in the back pressure chamber **50** is adjusted to be higher than the pressure in the low-pressure chamber **20**. It should be noted that a pressure release hole **16** is formed in the inner casing **3**, penetrates the inner casing **3** in the radial direction, and allows the low-pressure chamber **20** and the back pressure chamber **50** to communicate with each other. In addition, a pressure adjustment valve **45** is provided in the pressure release hole **16**. The pressure adjustment valve **45** is opened by the pressure of the back pressure chamber **50** exceeding a set value.

(35) In addition, the boss **42c** of the movable scroll **42** is inserted through a through hole **8b** in the middle of the thrust plate **8**. The through hole **8b** is formed to have a diameter size at which it is possible to allow eccentric rotation by the eccentric portion **2a** of the rotary shaft **2** fitted into the boss **42c**. In other words, the sliding surface **7a** of the side seal **7** is capable of sliding relative to the sliding surface **8a** of the thrust plate **8** with eccentric rotation by the eccentric rotation of the rotary shaft **2** (see FIG. 4).

(36) It should be noted that FIGS. 4A to 4D in FIG. 4 illustrate the rotational trajectory of the boss **42c** that is viewed from the fixed scroll **41** side and indicated by the black arrows. Respectively illustrated in FIGS. 4B to 4D are the boss **42c** rotated by 90 degrees, 180 degrees, and 270 degrees with FIG. 4A serving as the clockwise reference. In addition, the sliding region between the sliding surface **7a** of the side seal **7** and the sliding surface **8a** of the thrust plate **8** is schematically illustrated by dots. In addition, regarding the rotary shaft **2**, the counterweight portion **2b** constituting the eccentric mechanism and so on are not illustrated and only the eccentric portion **2a** fitted into the boss **42c** is illustrated for convenience of description.

(37) As described above, the side seal **7** is a sliding component having the sliding surface **7a** sliding relative to the sliding surface **8a** of the thrust plate **8** with the eccentric rotation.

(38) Next, dynamic pressure generation during the sliding of the side seal **7** relative to the thrust plate **8** will be described with reference to FIG. 5. It should be noted that FIG. 5 illustrates an aspect when the side seal **7** moves from the state of FIG. 4D toward the state of FIG. 4A. In addition, in FIG. 5, the side seal **7** in a case where the sliding surface **7a** is viewed from the axial direction is illustrated, and the circle marks illustrated in the enlarged portions indicate the points where the pressure increases in the dynamic pressure generation groove **70**. In addition, a fluid containing, for example, a refrigerant and lubricating oil is stored in the dynamic pressure generation groove **70** even when the rotation is stopped.

(39) As illustrated in FIG. 5, when the side seal **7** is to move in the direction of the white arrow, the fluid in the dynamic pressure generation groove **70** moves in the opposite direction relative to the white arrow. As a result, dynamic pressure is generated on the inner side wall **70a** and the outer side wall **70b** partitioning the dynamic pressure generation groove **70**, the sliding surfaces **7a** and **8a** are slightly separated from each other, and a fluid film is formed by the fluid.

(40) Specifically, when the thrust plate **8** is regarded as an analog clock, the position directly above the paper surface is set to the 12 o'clock position and, at the 12 o'clock position in the dynamic pressure generation groove **70**, dynamic pressure is mainly generated on each intersection surface **701** on the inner side wall **70a**. In addition, at the 3 o'clock position in the dynamic pressure generation groove **70**, dynamic pressure is mainly generated on each intersection surface **704** on the outer side wall **70b**. In addition, at the 6 o'clock position in the dynamic pressure generation groove **70**, dynamic pressure is mainly generated on each intersection surface **703** on the outer side wall **70b**. In addition, at the 9 o'clock position in the dynamic pressure generation groove **70**, dynamic pressure is mainly generated on each intersection surface **702** on the inner side wall **70a**.

(41) As described above, the inner side wall **70a** and the outer side wall **70b** of the dynamic pressure generation groove **70** extend in the circumferential direction in a waveshape and have the plurality of intersection surfaces **701** to **704** intersecting with respect to the eccentric rotation of the side seal **7**. Accordingly, dynamic pressure can be generated by the intersection surfaces **701** to **704** regardless of the eccentric rotation angle of the side seal **7**, the sliding surfaces **7a** and **8a** can be separated from each other with the inclination of the sliding surfaces **7a** and **8a** reduced, and the frictional resistance of the sliding surfaces **7a** and **8a** can be stably reduced.

(42) In addition, at the 12 o'clock and 6 o'clock positions in the dynamic pressure generation groove **70**, the intersection surfaces **701** and **703** have a concave shape holding the fluid in the dynamic pressure generation groove **70** that moves toward the opposite direction relative to the white arrow. Accordingly, the fluid near the intersection surfaces **701** and **703** is unlikely to be dispersed in the circumferential direction, and dynamic pressure is generated with ease. In other words, a larger dynamic pressure is generated at the 12 o'clock and 6 o'clock positions in the dynamic pressure generation groove **70** than at the 3 o'clock and 9 o'clock positions in the dynamic pressure generation groove **70**.

(43) It should be noted that FIG. 5 illustrates a form when the side seal **7** moves from the state of FIG. 4D toward the state of FIG. 4A, dynamic pressure is generated in substantially the same form when the side seal **7** moves from the state of FIG. 4D toward the state of FIG. 4A, when the side seal **7** moves from the state of FIG. 4A toward the state of FIG. 4B, when the side seal **7** moves from the state of FIG. 4B toward the state of FIG. 4C, and when the side seal **7** moves from the state of FIG. 4C toward the state of FIG. 4D, and thus description is omitted as to the forms when the side seal **7** moves from the state of FIG. 4A toward the state of FIG. 4B, from the state of FIG. 4B toward the state of FIG. 4C, and from the state of FIG. 4C toward the state of FIG. 4D.

(44) In addition, in the sliding of the side seal **7** relative to the thrust plate **8** and entailing eccentric rotation, the position where a large dynamic pressure is generated in the dynamic pressure generation groove **70** continuously moves along the circumferential direction of the dynamic

pressure generation groove **70** in accordance with the movement direction of the sliding surface **7a** of the side seal **7**. Accordingly, a state where the relative inclination between the sliding surfaces **7a** and **8a** is suppressed can be maintained.

(45) In addition, the dynamic pressure generation groove **70** is formed in an annular shape, and thus dynamic pressure can be generated over the entire circumference of the sliding surface **7a** and the inclination of the sliding surfaces **7a** and **8a** can be reduced regardless of the eccentric rotation angle of the side seal **7**.

(46) In addition, the inner side wall **70a** and the outer side wall **70b** have a waveshape with a constant period, and thus dynamic pressure can be equally generated over the circumferential direction of the sliding surface **7a**.

(47) In addition, the inner side wall **70a** and the outer side wall **70b** have a waveshape having a curved surface shape, and thus the movement of the fluid in the dynamic pressure generation groove **70** becomes smooth and dynamic pressure can be stably generated.

(48) In addition, the back pressure chamber **50** extends to the inner diameter side of the sliding surfaces **7a** and **8a**. Accordingly, when the sliding surfaces **7a** and **8a** are separated from each other, the fluid in the back pressure chamber **50** is introduced from the inner diameter side of the sliding surfaces **7a** and **8a**. In addition, when the scroll compression mechanism **4** is driven, the pressure of the back pressure chamber **50** increases, a high-pressure fluid is introduced between the sliding surfaces **7a** and **8a** from the back pressure chamber **50**, and thus the sliding surfaces **7a** and **8a** can be further separated from each other by the pressure of the fluid.

(49) In addition, although the present embodiment exemplifies a form in which the radial width dimension **L5** (that is, the separation width between the inner side wall **70a** and the outer side wall **70b**) of the dynamic pressure generation groove **70** is constant over the entire circumference, the present invention is not limited thereto. In an alternative disposition, the radial width dimension of the dynamic pressure generation groove may be different at different positions in the circumferential direction of the dynamic pressure generation groove. For example, the inner side wall and the outer side wall may be formed in waveshapes with different amplitudes.

Second Embodiment

(50) Next, a dynamic pressure generation groove **170** of a side seal **107** as a sliding component according to a second embodiment of the present invention will be described with reference to FIG. **6**. It should be noted that the description of configurations identical to those of the first embodiment will be omitted for redundancy avoidance.

(51) As for the side seal **107** in the second embodiment illustrated in FIG. **6**, a plurality of the dynamic pressure generation grooves **170** extending in an arc shape in the circumferential direction are disposed apart from each other in the circumferential direction of a sliding surface **107a**.

(52) According to this, dynamic pressure can be generated in both circumferential end portions **170d** and **170e** of each dynamic pressure generation groove **170** in addition to an inner side wall **170a** and an outer side wall **170b** of each dynamic pressure generation groove **170**. It should be noted that in this case, it is preferable that the dynamic pressure generation grooves **170** are equally arranged in the circumferential direction.

Third Embodiment

(53) Next, a dynamic pressure generation groove **270** of a side seal **207** as a sliding component according to a third embodiment of the present invention will be described with reference to FIG. **7**. It should be noted that the description of configurations identical to those of the first embodiment will be omitted for redundancy avoidance.

(54) As for the dynamic pressure generation groove **270** of the side seal **207** in the third embodiment illustrated in FIG. **7**, an inner side wall **270a** is wavy when viewed from the axial direction. In addition, an outer side wall **270b** is circular when viewed from the axial direction. In this manner, the dynamic pressure generation groove may be wavy on at least one of the inner and outer side walls.

(55) In addition, the inner side wall **270a** has an outer diameter side apex **275** that tapers with an acute angle. In addition, the inner side wall **270a** has a waveshape. According to this, the circumferential width of the outer diameter side apex **275** can be reduced, and thus it is possible to ensure a large intersection surface **271** that has a circular arc shape, is convex to the inner diameter side, and is formed between the outer diameter side apexes **275** adjacent to each other.

(56) Although embodiments of the present invention have been described above with reference to the drawings, the specific configurations are not limited to the embodiments and any changes or additions within the scope of the present invention are included in the present invention.

(57) For example, although a form in which the inner and outer side walls have a waveshape having a curved surface shape has been exemplified in the first to third embodiments, the present invention is not limited thereto and the waveshape may be, for example, rectangular or triangular when viewed from the axial direction.

(58) In addition, although an aspect in which the side seal **7** as a sliding component is applied to the scroll compressor **C** used in the air conditioning system of an automobile or the like has been described in the first to third embodiments, the present invention is not limited thereto and the side seal **7** as a sliding component may be applied to, for example, a scroll expansion compressor provided integrally with an expander and a compressor insofar as it is a rotating machine including an eccentric mechanism.

(59) In addition, each of the fluids in the spaces inside and outside the sliding surface of the sliding component may be any of a gas, a liquid, and a gas-liquid mixture.

(60) In addition, insofar as the sliding component of the present invention has a sliding surface that relatively slides with eccentric rotation, the sliding component of the present invention may be used in an environment in which the pressure inside the sliding surface and the pressure outside the sliding surface are equal to each other without being limited to an environment in which there is a pressure difference between the inside and outside of the sliding surface. In addition, the sliding component of the present invention does not have to function as a seal and may be one capable of stably reducing the frictional resistance of a sliding surface.

(61) In addition, although the side seal having the relatively sliding surface is made of resin and the thrust plate is made of metal in the first to third embodiments, the material of the sliding component may be freely selected in accordance with the environment of use and so on.

(62) In addition, although an aspect in which a dynamic pressure generation groove is formed in the sliding surface of the side seal has been described in the first to third embodiments, the present invention is not limited thereto. A dynamic pressure generation groove may be formed in the sliding region of the sliding surface of the thrust plate (see FIG. **4**), which is a sliding component having a sliding surface relatively sliding with eccentric rotation. In addition, dynamic pressure generation grooves may be formed in both the sliding surface of the side seal and the sliding surface of the thrust plate.

(63) In addition, although a configuration in which the sliding surface of the side seal and the sliding surface of the thrust plate as a sliding component slide relative to each other with eccentric rotation has been described in the first to third embodiments, the present invention is not limited thereto. A dynamic pressure generation groove may be formed in the sliding surface relatively sliding with eccentric rotation with only one of the side seal and the thrust plate provided. For example, in a case where only the thrust plate is provided, dynamic pressure generation grooves may be formed in one or both of the sliding surface of the thrust plate as a sliding component and the back surface of the end plate of the movable scroll. In addition, in a case where only the side seal is provided, a dynamic pressure generation groove may be formed in the sliding surface of the side seal as a sliding component. In this case, the side seal also functions as a thrust bearing that abuts against the inner peripheral surface of the inner casing and receives the axial load of the movable scroll.

(64) In addition, in a case where the side seal and the thrust plate are not provided and the back

surface of the end plate of the movable scroll functions as a thrust bearing that abuts against the inner peripheral surface of the inner casing and receives the axial load of the movable scroll, a dynamic pressure generation groove may be formed in the sliding surface formed on the back surface of the end plate of the movable scroll.

(65) In addition, the sliding surface may be provided with a conduction groove for conduction between the external space on the inner diameter side or the outer diameter side of the sliding surface and the dynamic pressure generation groove.

(66) In addition, although a form in which the external space on the low-pressure side is on the outer diameter side of the side seal and the high-pressure external space is on the inner diameter side of the side seal has been exemplified, the external space on the low-pressure side may be on the inner diameter side of the side seal with the high-pressure external space on the outer diameter side of the side seal.

REFERENCE SIGNS LIST

(67) **4** Scroll compression mechanism **7** Side seal (sliding component) **7a** Sliding surface **8** Thrust plate **8a** Sliding surface **20** Low-pressure chamber (external space on outer diameter side) **30** High-pressure chamber **40** Compression chamber **41** Fixed scroll **42** Movable scroll **50** Back pressure chamber (external space on inner diameter side, external space on high-pressure side) **70** Dynamic pressure generation groove **70a** Inner side wall (side wall on inner diameter side) **70b** Outer side wall (side wall on outer diameter side) **107** Side seal (sliding component) **107a** Sliding surface **170** Dynamic pressure generation groove **170a** Inner side wall (side wall on inner diameter side) **170b** Outer side wall (side wall on outer diameter side) **207** Side seal (sliding component) **270a** Inner side wall (side wall on inner diameter side) **270b** Outer side wall (side wall on outer diameter side) **271** Intersection surface **701** to **704** Intersection surface C Scroll compressor M Drive motor

Claims

1. An eccentric sliding assembly comprising: a first sliding component having a sliding surface; a second sliding component having a sliding surface slidable to the sliding surface of the first sliding component; a rotational member configured to rotate around a predetermined axis; a movable member which is the first sliding component or to which the first sliding component is attached; and a fixed member which is the second sliding component or to which the second sliding component is attached; wherein the rotational member has an eccentric portion eccentrically formed with respect to the predetermined axis and protruded toward the movable member, the movable member has a boss fitted around the eccentric portion of the rotational member and is relatively rotatable therewith, and the sliding surface of at least one of the first sliding component and the second sliding component is provided with a dynamic pressure generation groove defined by side walls extending in a circumferential direction, at least one of the side walls being formed in a waveshape with amplitude in a radial direction.

2. The eccentric sliding assembly according to claim 1, wherein both the side walls of the dynamic pressure generation groove are formed in the waveshape with the amplitude in the radial direction.

3. The eccentric sliding assembly according to claim 1, wherein the dynamic pressure generation groove is formed in an annular shape.

4. The eccentric sliding assembly according to claim 1, wherein the waveshape of the side wall is periodically formed with a constant pitch.

5. The eccentric sliding assembly according to claim 1, wherein the waveshape of the side wall is formed by curved surfaces.

6. The eccentric sliding assembly according to claim 2, wherein the dynamic pressure generation groove is formed in an annular shape.

7. The eccentric sliding assembly according to claim 2, wherein the waveshape of the side wall is periodically formed with a constant pitch.

8. The eccentric sliding assembly according to claim 2, wherein the waveshape of the side wall is formed by curved surfaces.
 9. The eccentric sliding assembly according to claim 3, wherein the waveshape of the side wall is periodically formed with a constant pitch.
 10. The eccentric sliding assembly according to claim 3, wherein the waveshape of the side wall is formed by curved surfaces.
 11. The eccentric sliding assembly according to claim 4, wherein the waveshape of the side wall is formed by curved surfaces.
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